

Week 4: Work and Kinetic Energy – Supplementary Notes

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These notes support the Week 4 lecture. They contain background that was moved out of the contact time and are not needed for the lecture problems.

1 The dot product, briefly

For vectors $\vec{A}$ and $\vec{B}$ with angle ϕ between them (tail to tail), $\vec{A} \cdot \vec{B} = AB \cos \phi = A_x B_x + A_y B_y + A_z B_z$. It is positive for $\phi < 90^\circ$, zero for $\phi = 90^\circ$ and negative for $\phi > 90^\circ$. For work, $W = \vec{F} \cdot \vec{s}$.

2 A short calculus refresher

$\int x^n dx = \frac{x^{n+1}}{n+1}$ for $n \neq -1$. Definite integral: $\int_a^b x^n dx = \left[\frac{x^{n+1}}{n+1} \right]_a^b$. Example: $\int_0^2 3x^2 dx = [x^3]_0^2 = 8$.

8. For a constant factor, $\int c f dx = c \int f dx$.

3 The work–energy theorem for a varying force

For motion along x with a force $F_x(x)$, Newton's second law gives $F_x = ma_x = m \frac{dv_x}{dt}$. Since $dx = v_x dt$,

$$W_{\text{tot}} = \int_{x_1}^{x_2} F_x dx = \int_{x_1}^{x_2} m \frac{dv_x}{dt} dx = \int_{v_1}^{v_2} m v_x dv_x = \frac{1}{2} m v_2^2 - \frac{1}{2} m v_1^2.$$

This shows the theorem holds whatever the force, which is why we could use it with a spring.

4 Work along a general path

For a force that varies along a curved path, the work is the line integral

$$W = \int_{\text{path}} \vec{F} \cdot d\vec{r},$$

the sum over tiny displacements $d\vec{r}$ of $F dr \cos \phi$ with the angle ϕ changing along the path. A force perpendicular to $d\vec{r}$ at every point (for example, tension in a pendulum string) contributes zero.

5 Where does the energy lost to friction go?

Kinetic friction does negative work on a sliding object and reduces its kinetic energy. The energy is not destroyed: it appears as the *thermal energy* of the two surfaces (they warm up). The work–energy theorem as used in this lecture counts only the kinetic energy of the object, so friction appears as negative work. The full accounting, with potential energy and thermal energy, is the subject of the next lecture and of the thermodynamics part of the course.

6 Units and conversions

$1 \text{ J} = 1 \text{ N m}$; $1 \text{ W} = 1 \text{ J/s}$; $1 \text{ kW} = 1000 \text{ W}$; $1 \text{ hp} = 746 \text{ W}$; $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$. A person climbing stairs develops roughly 10^2 W ; a car cruising on a road uses roughly 10^4 W .